

ARM Power!

Comparing MySQL x86 vs ARM Performance

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About Me

- Support Engineer at Percona since 2017.
- Working with MySQL for over nine years.
- Working with databases for over twelve years.
- Speaker at conferences and meetups about MySQL/MongoDB.
- [LinkedIn](#) (Vinicius Grippa)



What is ARM?

What is ARM and its Differences to Intel/AMD(x86)?

- Advanced RISC Machines (ARM).
- RISC machines are composed of small, highly optimized set of instructions.
- Power efficiency.
- ARM Processor is affordable, hence the reason it is present in lots of mobiles and in the general industry.

Example

- Apple M1 chip is based on ARMv8.6-A architecture.
- Graviton AWS instances are based on ARM.

What is ARM and its differences to Intel/AMD(x86)

- X86 machines are CISC machines.
- CISC = complex instruction set computer .
- Combining simple instructions into a single complex one, thereby reducing the number of instructions.
- Simplifying a software or application depends on the complexity of the hardware. To simplify an application, usually the hardware needs to be complex.

Example

- Intel
- AMD

Benchmark

AWS Instances and Operating System

- **For x86**
 - m5d.8xlarge (128 GiB of Memory, 32 vCPUs, 2 x 600 NVMe SSD, 64-bit platform)
 - CentOS Linux release 7.9.2009 (Core)
- **For ARM**
 - m6g.8xlarge (128 GiB of Memory, 32 vCPUs, 1 x 1900 NVMe SSD)
 - CentOS Linux release 7.9.2009 (AltArch)

AWS Instances Cost

x86

Instance name ▲	On-Demand hourly rate ▼	vCPU ▼	Memory ▼	Storage ▼	Network performance ▼
m5d.8xlarge	\$1.808	32	128 GiB	2 x 600 NVMe SSD	10 Gigabit

ARM

Instance name ▲	On-Demand hourly rate ▼	vCPU ▼	Memory ▼	Storage ▼	Network performance ▼
m6gd.8xlarge	\$1.4464	32	128 GiB	1 x 1900 NVMe SSD	10 Gigabit

Cost Comparison

- ARM is 25% cheaper than x86
- 1-month
 - x86 cost: \$1319,84 USD
 - ARM cost: \$1055,58 USD
- \$264,26/month = \$ 3168 USD/year

- x86 1-CPU=\$41,24 USD/month
- ARM 1-CPU=\$32,98 USD/month

MySQL

Configuration

The my.cnf configuration

```
innodb_buffer_pool_size = 100G
innodb_log_file_size = 10G
relay-log-index=mysql-relay
relay-log=mysql-relay
log-bin=mysql-bin
sync_binlog = 0
```

Version

x86

```
# mysqld --version
/usr/sbin/mysqld Ver 8.0.23 for Linux on x86_64 (MySQL Community Server
- GPL)
```

ARM

```
$ mysqld --version
/usr/sbin/mysqld Ver 8.0.23 for Linux on aarch64 (MySQL Community
Server - GPL)
```

Sysbench

- Used an isolated server for testing
c5a.8xlarge (32 vCPUs, 64 GiB)

- TPCC-Like Workload

- Preparing the data

```
/usr/share/sysbench/tpcc.lua --mysql-host=3.236.47.139  
--mysql-port=3306 --mysql-user=sysbench --mysql-password=sekret  
--mysql-db=percona --time=300 --threads=8 --report-interval=1  
--tables=10 --scale=100 --db-driver=mysql prepare
```

- Run tests with 8,16,32,64,128,256 threads

```
/usr/share/sysbench/tpcc.lua --mysql-host=172.30.155.116  
--mysql-port=3306 --mysql-user=sysbench --mysql-password=sekret  
--mysql-db=percona --time=300 --threads=256 --report-interval=1  
--tables=10 --scale=100 --db-driver=mysql run
```

TPCC Results

Results

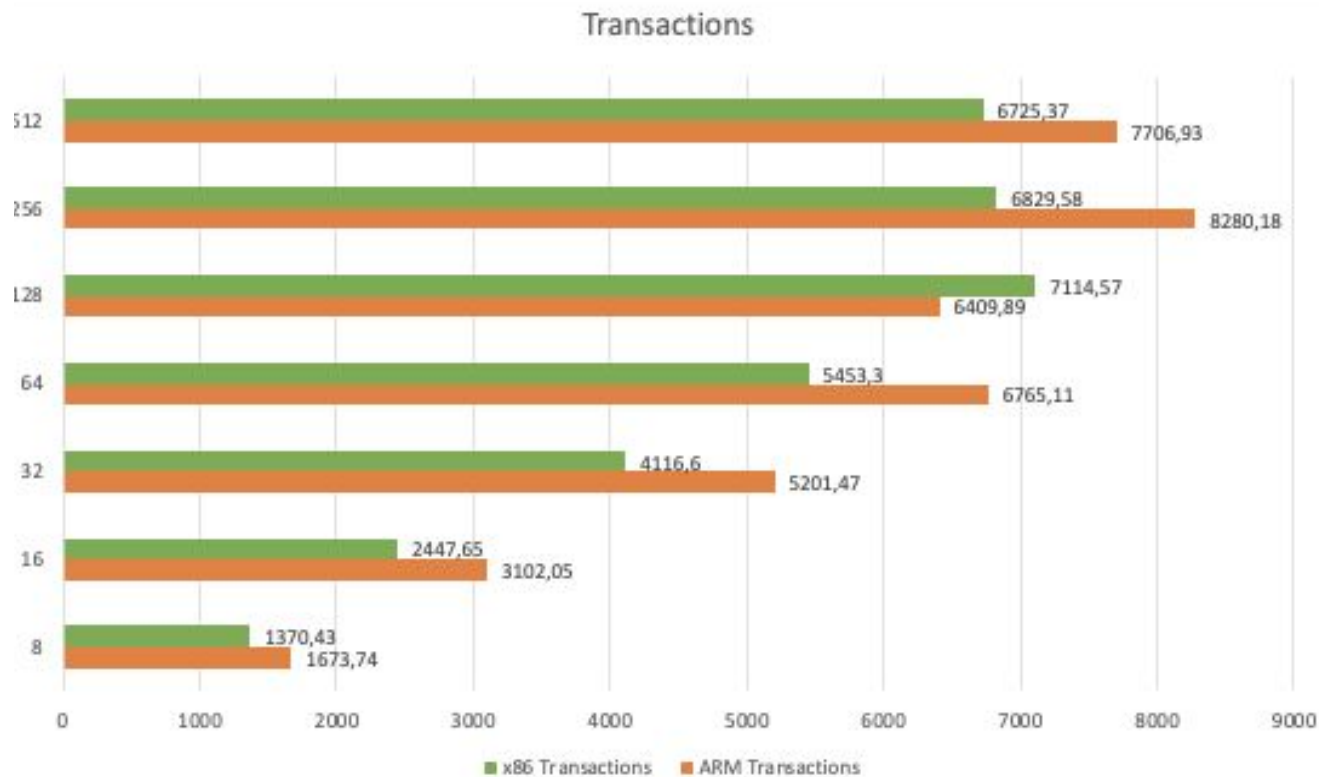
ARM

Threads	Transactions	Queries
8	1673,74	47560,16
16	3102,05	88183,16
32	5201,47	147874,72
64	6765,11	192430,65
128	6409,89	182445,37
256	8280,18	235371,77
512	7706,93	219236,24

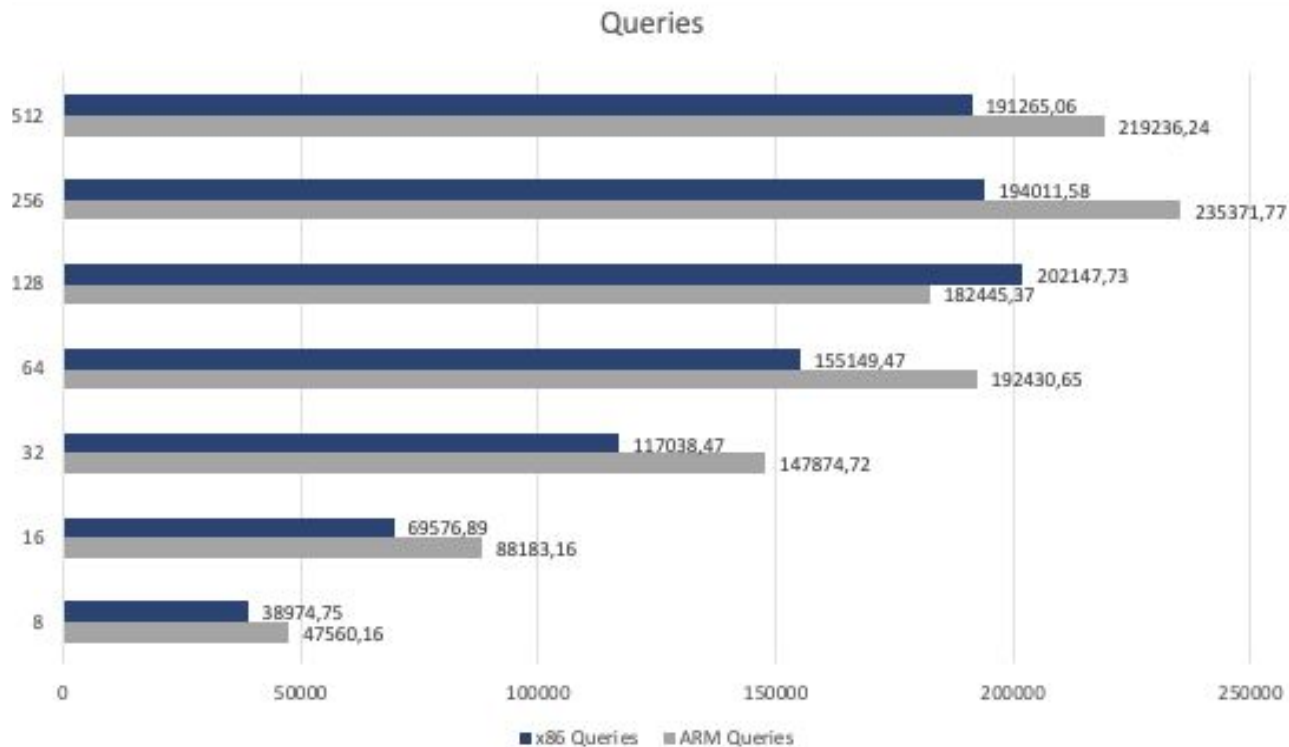
x86

Threads	Transactions	Queries
8	1370,43	38974,75
16	2447,65	69576,89
32	4116,6	117038,47
64	5453,3	155149,47
128	7114,57	202147,73
256	6829,58	194011,58
512	6725,37	191265,06

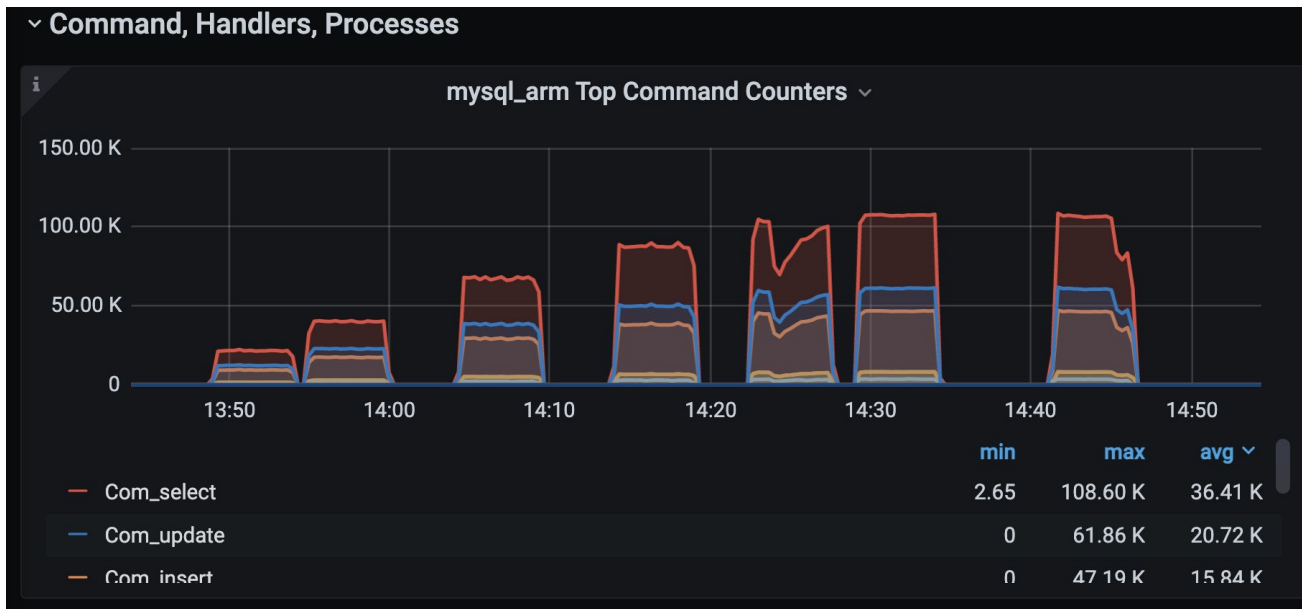
Results



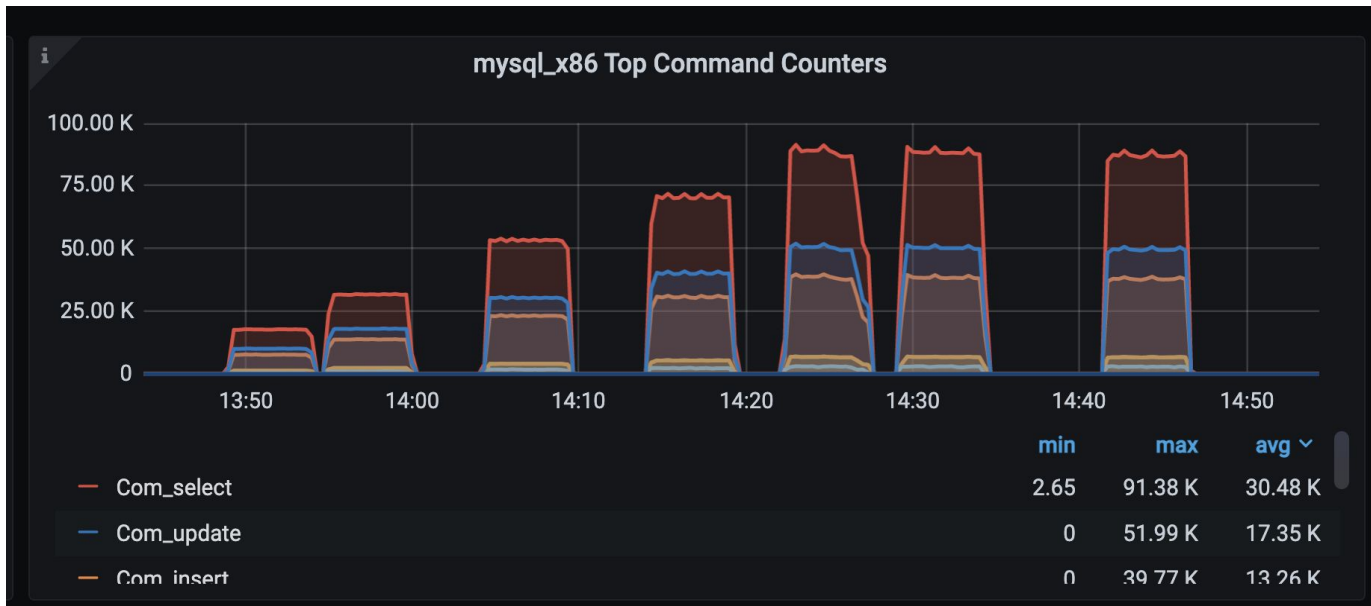
Results



Results



Results



OLTP Results

Results - What About Read-Only?

ARM

Threads	Transactions	Queries
8	3468,42	55494,7
16	6607,66	105722,58
32	12016,89	192270,16
64	16029,56	256472,9
128	20838,06	333408,93
256	21316,54	341064,72
512		

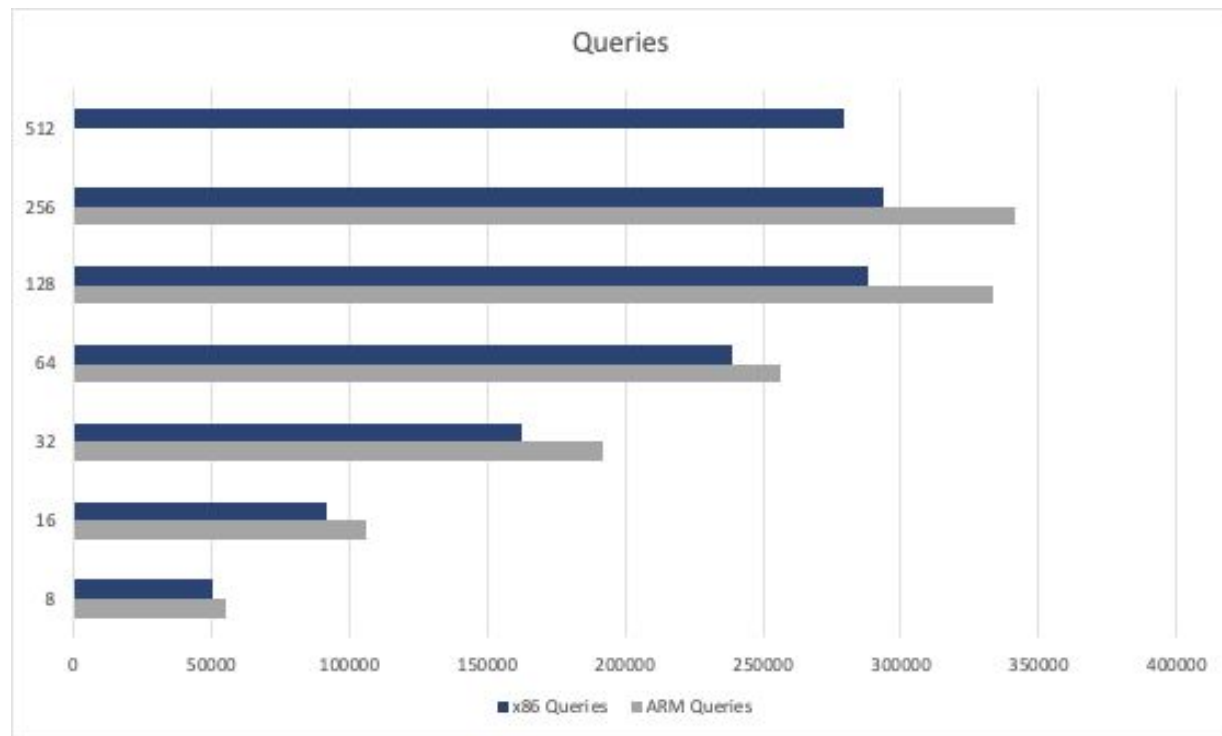
x86

Threads	Transactions	Queries
8	3134,5	50152,04
16	5743,48	91895,66
32	10159,43	162550,82
64	14947,43	239158,88
128	17991,71	287867,28
256	18329,28	293268,47
512	17480,51	279688,18

Results - What About Read-Only?

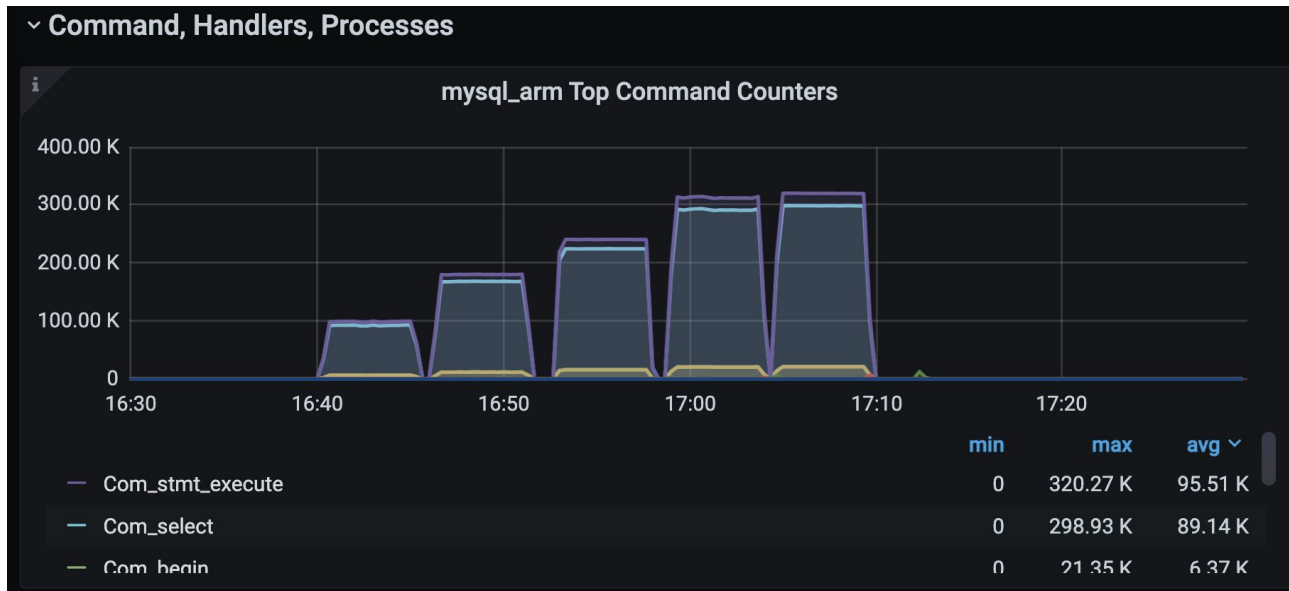


Results - What About Read-Only?



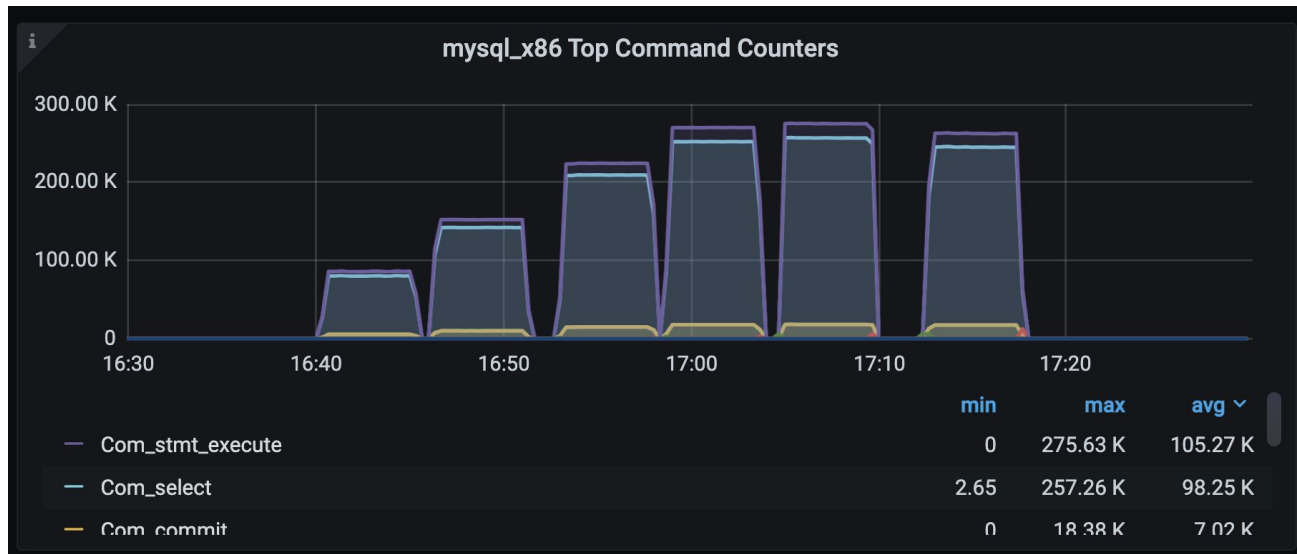
Results - What About Read-Only?

ARM



Results - What About Read-Only?

x86



Problems

MySQL ARM Instance Run Out of Memory

```
# free -m  
bash: fork: Cannot allocate memory
```

What About the Ecosystem?

The Percona Ecosystem

Challenges

- We need to compile Percona Xtrabackup (tested with Ubuntu and Centos)
- We need to compile Percona PMM agent (tested with CentOS)

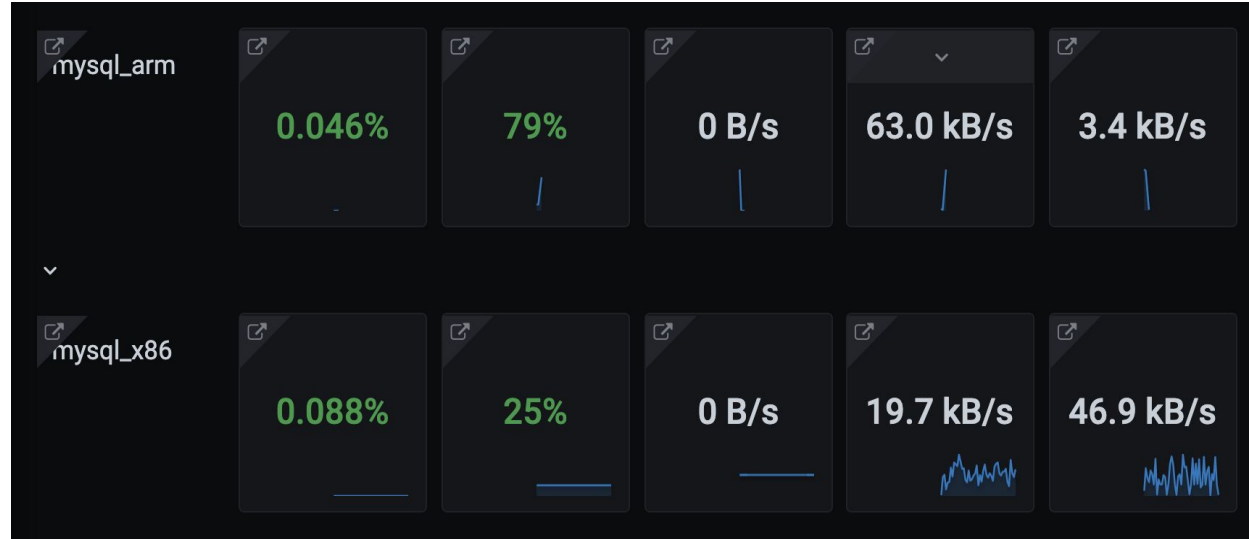
Xtrabackup

```
# /pxb/percona-xtrabackup/build/bin/xtrabackup --version
xtrabackup: recognized server arguments: --datadir=/mysql_data/data
--innodb_buffer_pool_size=80G --innodb_log_file_size=10G
--log_bin=mysql-bin
/pxb/percona-xtrabackup/build/bin/xtrabackup version 8.0.23-16 based on
MySQL server 8.0.23 Linux (aarch64) (revision id: 934bc8f)
```

To compile PXB:

<https://www.percona.com/blog/2021/04/28/compiling-percona-xtrabackup-for-arm/>

PMM



PMM

Compiling pmm-agent is not simple and not recommended for Production.

Using the remote add instance option is the best alternative in the moment.

PMM



Conclusion

Conclusion

ARM beat x86 in both workloads being ~25% cheaper

TPCC

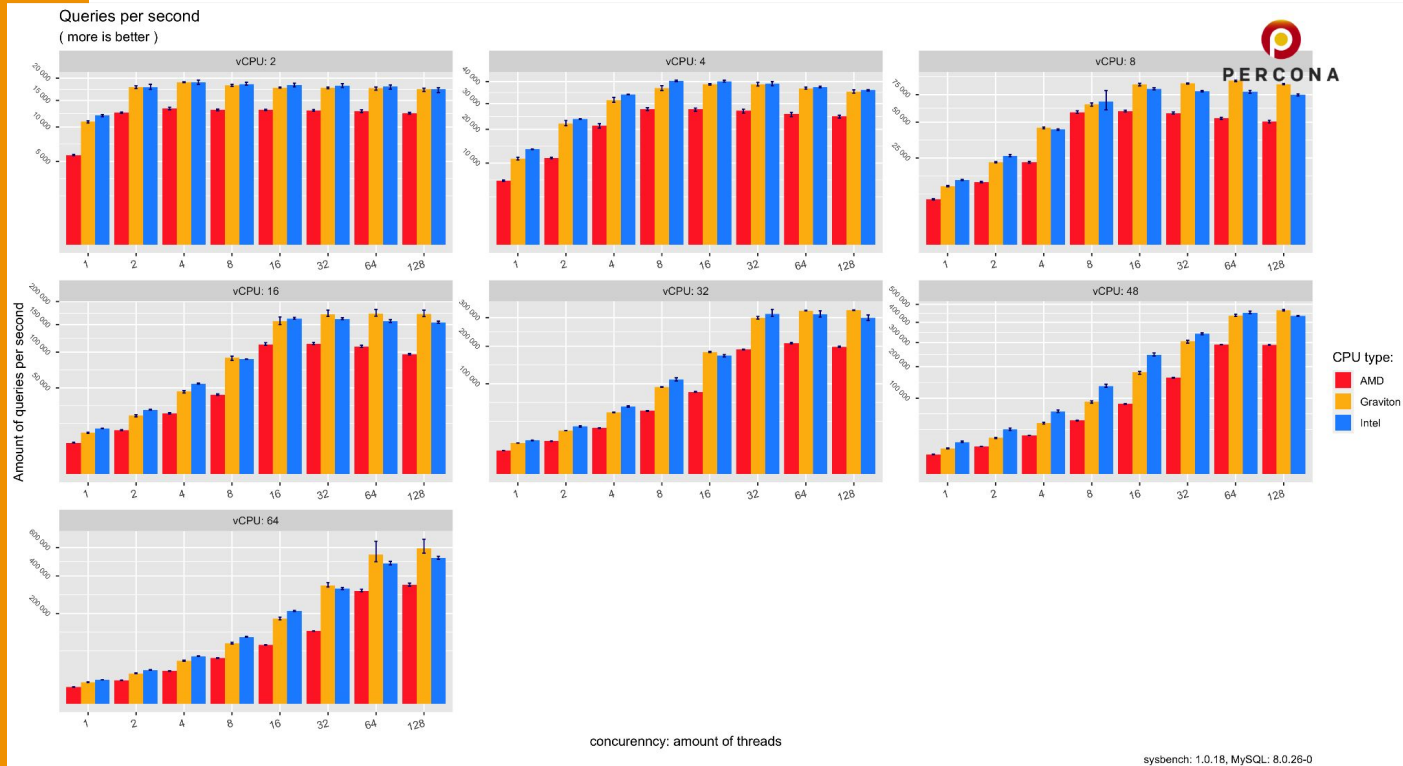
Threads	% Queries (x86/ARM)	% Transactions (x86/ARM)
8	18%	18%
16	21%	21%
32	21%	21%
64	19%	19%
128	-11%	-11%
256	18%	18%
512	13%	13%

Conclusion

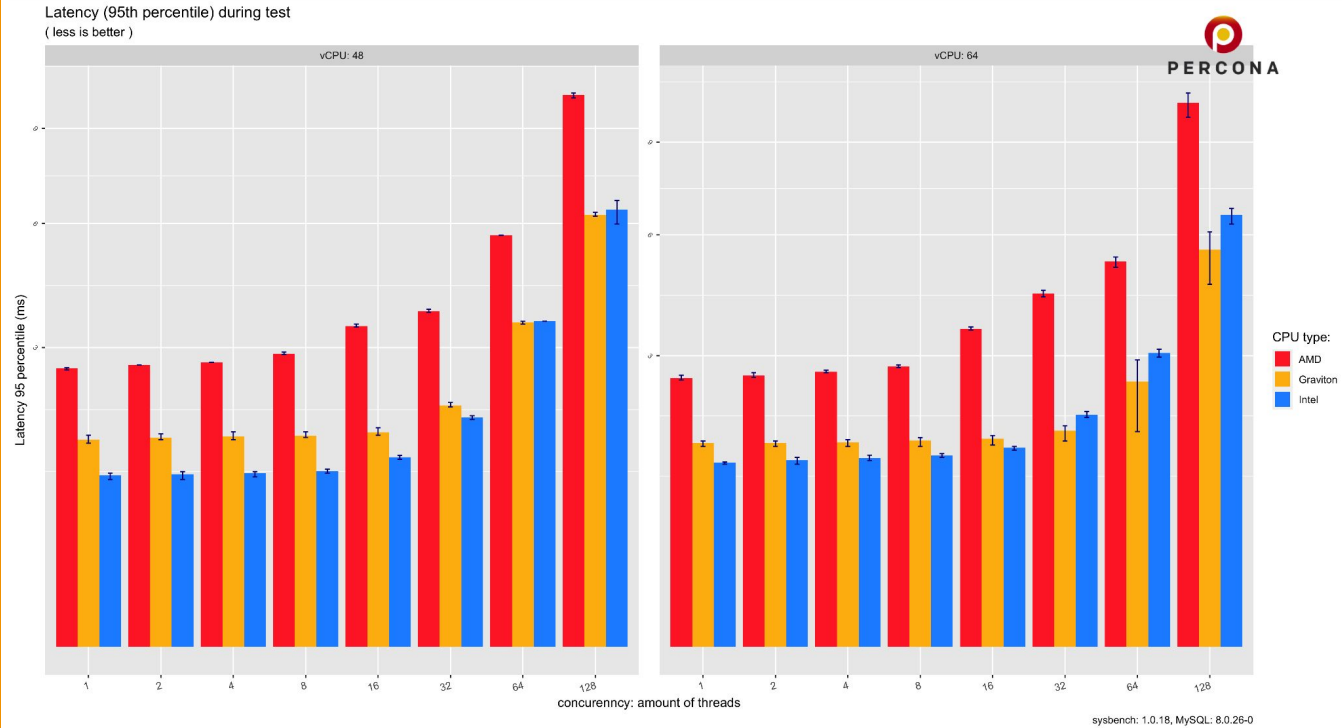
Read-Only

Threads	% Queries (x86/ARM)	% Transactions (x86/ARM)
8	10%	10%
16	13%	13%
32	15%	15%
64	7%	7%
128	14%	14%
256	14%	14%
512		

Not the entire story...



One more...



Conclusion

- **ARM** is growing and some tools are necessary to compile.
- It does not have default ARM packages for Ubuntu/Debian.
- **x86** has been in the market for years, it is much more stable and reliable.
- We can find a greater number of tools available for x86.
- To do a fair comparison, both processors should be the same number of cores and similar release date.

References

- https://github.com/guriandoro/pmm_arm (Thanks Agustin with the help!)
- <https://www.percona.com/blog/2020/08/04/compiling-a-percona-monitoring-and-management-v2-client-in-arm-architecture/>
- <https://mysqlonarm.github.io/> (Highlights to Krunal)
- <https://www.percona.com/blog/2021/04/28/compiling-percona-xtra-backup-for-arm/>

PostgreSQL x86 x ARM

- <https://www.percona.com/blog/2021/01/22/postgresql-on-arm-based-aws-ec2-instances-is-it-any-good/>

谢谢

Thank you

Grazie

Obrigado

Gracias

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